

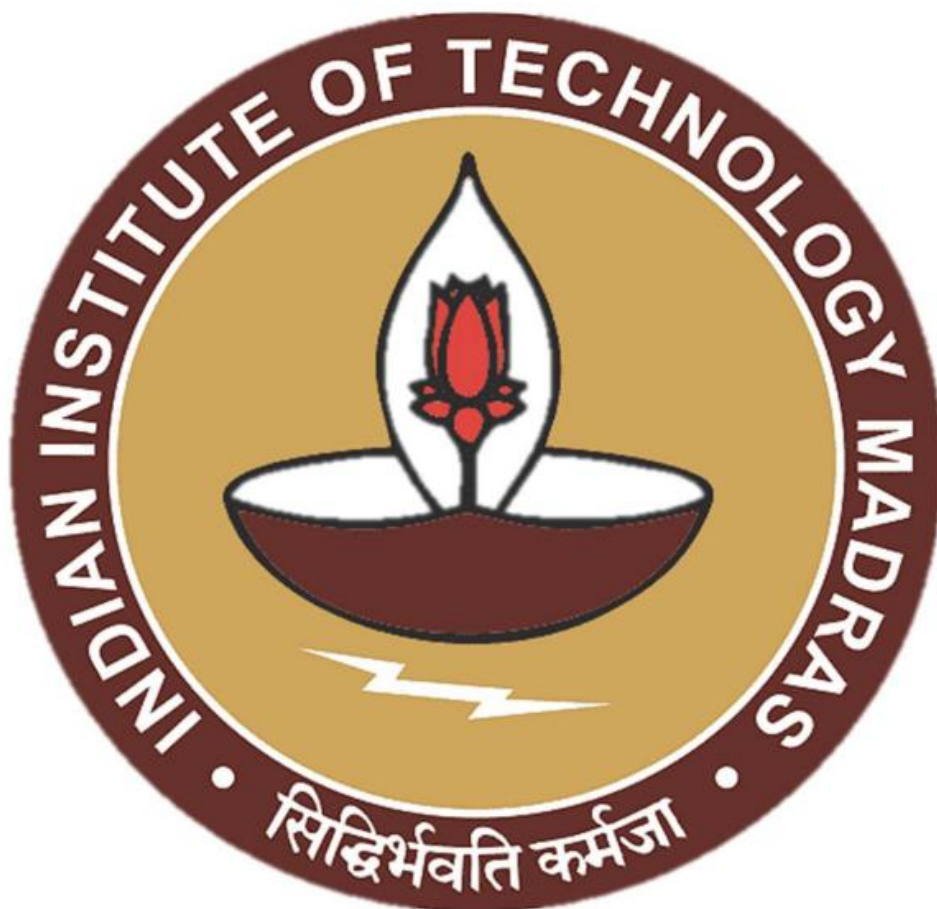
Optimizing Small F&B businesses with Data Analytics

A Final-Term Report for BDM Capstone Project

Submitted by

Name: Piyush Jha

Roll Number: 21F1002373



IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Contents

1. Executive Summary	1
2. Detailed Explanation of Analysis Process/Methods	2-7
3. Results and Findings	7-15
4. Interpretation of Results and Recommendations	16-18

1. Executive Summary

This project, titled “*Optimizing Small F&B Businesses with Data Analytics*”, focuses on **Shubham Sweet Corner**, a small food and beverage shop located at Lakshminiya Chowk, Polytechnic Bypass Road, Saharsa, Bihar. Despite its popularity among local residents for snacks and sweets, the business faces three persistent challenges — lack of systematic record-keeping, frequent employee turnover, and limited customer reach due to its distance from the nearby college area. These challenges collectively impact profitability, operational stability, and long-term growth.

To study these issues in a structured manner, **data was manually collected for 87 operational days between 15 June 2025 and 30 September 2025**. The dataset consisted of four components — *Product Pricing*, *Daily Sales Log*, *Employee Details*, and *Variable Expenses*. Analytical techniques such as **descriptive statistics, SKU-level profit analysis, and correlation checks** were applied using *Python (pandas, matplotlib)* and *Microsoft Excel*. Key business metrics including mean, median, standard deviation, and profit margins were computed to gain insights into sales patterns and revenue stability.

The analysis revealed that the shop’s average daily revenue is ₹10,333.77 with an average daily profit of ₹2,024.38 and an overall profit margin of 19.59%. The top five products contribute over 78% of total revenue, led by samosas with high volume but low margin.

To address the identified gaps, feasible low-cost interventions such as structured record-keeping, refundable employee deposits, SKU-level demand tracking, and a pushcart-based outreach strategy were proposed. These insights and actions together aim to create a more efficient, profitable, and data-informed small business model.

2. Detailed Explanation of Analysis Process/Method:

The analysis for *Shubham Sweet Corner* was conducted using a structured, step-by-step approach entirely in **Python**, with **Microsoft Excel** used only as a data entry and storage tool. The objective was to transform manually recorded business data into an analyzable format, perform descriptive and exploratory analytics, and extract insights to improve decision-making.

The workflow involved the following major stages:

- (a) Data Collection and Organization.
- (b) Data Cleaning and Pre-processing.
- (c) Descriptive Statistical Analysis.
- (d) Correlation and Trend Analysis.
- (e) Visualization and Pattern Identification.
- (f) Profitability Analysis.

a) Data Collection and Organization

Data was collected manually between **15 June 2025 and 30 September 2025**, representing **87 operational business days**. As the business had no digital system, data was entered daily into a structured Excel workbook under the supervision of the shop owner.

The workbook contained four distinct datasets:

1. **Product Pricing Sheet** – Listed 18 SKUs with the following columns: *Product Name*, *Cost Price (₹)*, *Selling Price (₹)*, *Profit per Unit (₹)*, and *Profit Margin (%)*. The *Cost Price* values already included the shop's total **employee cost** and **variable expenses** (gas, rent, and electricity). Hence, the *Profit per Unit* and *Profit Margin* in this sheet represented *net profit after all operational costs*.

Example:

- Samosa → Cost ₹9.4, Sell ₹10, Profit ₹0.6, Margin 6%
- Litti → Cost ₹6, Sell ₹10, Profit ₹4, Margin 40%
- Gulab Jamun → Cost ₹8.5, Sell ₹15, Profit ₹6.5, Margin 43.3%

2. **Daily Sales Log** – Contained SKU-wise daily quantities, selling prices, and total revenue for each product. Each SKU followed a repeating pattern of columns: **SKU_Qty, SKU_Selling_Price, SKU_Total_Revenue**. The dataset also included **Day, Date,** and **Total_Revenue_per_Day**, which represented the total sales revenue generated across all SKUs on a given date.
3. **Employee Details** – Contained names, roles, and monthly salaries of three staff members: Sonu, Vikas, and Mantu. This sheet was maintained for reference only, since employee cost was already embedded in SKU cost prices.
4. **Variable Expenses** – Included fixed monthly expenses such as Electricity ₹3,000, Gas ₹1,000, and Rent ₹8,000. These expenses were likewise already incorporated into the cost price of each product and not analyzed separately.

All sheets used consistent labelling and formatting, allowing for seamless import into Python using **pandas.read_excel()** for further cleaning and analysis.

b) Data Cleaning and Pre-processing

Given the manual nature of entry, cleaning and preprocessing were crucial before analysis. This step was performed completely in **Python** using the **pandas** library.

The following operations were carried out systematically:

1. Handling Missing or Incomplete Values:

The business remained closed between **20 July 2025 and 9 August 2025**.

For this interval, all records in the *Daily Sales Log* were set to NaN (Not a Number) to ensure these non-operational days were excluded from revenue calculations.

2. Standardizing Data Types:

- The *Date* column was converted to a datetime format to support chronological operations such as sorting and time-series visualization.
- All numeric fields (quantities, prices, and revenues) were coerced into numeric data types (float64) for computation.

3. Ensuring Structural Consistency:

All columns followed a uniform naming convention: <Product>_Qty, <Product>_Selling_Price, and <Product>_Total_Revenue.

This allowed aggregation functions to automatically identify and summarize values per SKU or per day.

4. Removing Redundant Columns:

The *Category* column in the Product Pricing sheet was empty and hence dropped.

No Excel formulas, functions, or macros were used at any point. All analysis, aggregation, and summary statistics were computed purely using Python.

c) Descriptive Statistical Analysis

Descriptive statistics were performed on the *Total_Revenue_per_Day* column to summarize daily business performance. Using `DataFrame.describe()` in Python, the following results were obtained:

Statistic	Value
Count	87
Mean	₹10,333.77
Standard Deviation	₹556.67
Median	₹10,397.00
Minimum	₹9,009.00
Maximum	₹13,432.00

These results show that the business maintained consistent daily performance, with limited fluctuation (a standard deviation of 5.3% of the mean).

Such stability reflects steady customer demand and strong baseline sales throughout the period.

The same analysis was extended to SKU-level sales to identify the most profitable and high-demand items. The findings indicated that Samosa was the highest-selling SKU, averaging 408 units per day, contributing **39.5% of total revenue**, though it yielded only **6% margin**.

d) Correlation and Trend Analysis

To extend the analysis beyond descriptive statistics, a **correlation study** was conducted to examine the relationship between the quantity of each product sold and the overall daily revenue. This helped identify which specific products most strongly influenced daily business performance. The analysis was performed in Python using the `pandas.corr()` function, with results visualized through a red-colored annotated heatmap and a focused bar chart of the top correlated products.

The correlation matrix revealed varying degrees of positive relationships between product quantities and total daily revenue. The **top five products most correlated with total revenue** were:

- **Paneer ($r = 0.62$)**
- **Gulab Jamun ($r = 0.54$)**
- **Rasgulla ($r = 0.53$)**
- **Laddoo ($r = 0.44$)**
- **Ashirwad Milk 1 Kg ($r = 0.40$)**

These results indicate that higher sales of paneer-based and sweet items are most closely associated with higher overall revenue, suggesting that these items significantly contribute to daily earnings.

e) Visualization and Pattern Identification

All visualizations were created using **Python's matplotlib and seaborn libraries** to clearly represent product-level profitability, sales trends, and correlation patterns derived from the dataset.

1. Profit per Unit per Product:

A horizontal bar chart was used to compare profit per unit across 18 SKUs. The chart, annotated with exact profit values, revealed that **Paneer, Laddoo and Dahi 1 Kg** had the highest per-unit profits, while **Samosa and Ashirwad Milk (1/2 Kg)** had the lowest. This highlighted the sharp contrast between high-margin specialty items and low-margin, high-volume staples.

2. Average Daily Revenue by Product:

A vertical bar chart illustrated each product's **average daily revenue** from June to September 2025, with values labeled above each bar. **Samosa** led in total revenue, followed by **Litti, Laddoo and Rasgulla**, confirming the business's dependence on volume-driven products rather than high-margin ones.

3. Daily Total Revenue with Closure Period:

A time-series line plot showed **Total Revenue per Day** across the observation period, with the closure gap (20 July–9 August 2025) shaded red. Revenue trends remained stable before and after the closure, peaking notably on weekends and on *15 August 2025*.

4. Correlation Heatmap and Top Correlated Products:

A red-themed annotated heatmap and bar chart displayed relationships between product quantities and daily revenue. The top correlated products were **Paneer ($r = 0.62$)**, **Gulab Jamun (0.54)**, **Rasgulla (0.53)**, **Laddoo (0.44)**, and **Ashirwad Milk (0.40)**, while **Samosa (-0.22)** showed a weak negative correlation, indicating high sales volume but low influence on overall revenue growth.

f) Profitability Analysis

Since the *Cost Price* of each SKU already included employee and variable expenses, the profit figures reflected **true net profitability per product**.

No additional cost allocation or adjustment was performed.

Using Python calculations, the **average profit per day** was computed as ₹2,024.38 — derived from **total daily revenue multiplied by the mean profit margin** of 19.59%. This implied that on a typical operational day, the business earned a net profit of roughly ₹2,000 after all expenses, indicating a sustainable model but limited scalability.

The profit-per-unit figures highlighted a key insight: although Samosa was the most sold item, its low margin (₹0.6 per unit) significantly restricted profit growth, suggesting the need for a price adjustment or bundled promotion strategy.

3. Results and Findings

This section presents the analytical outcomes obtained through Python-based statistical and visual methods. It summarizes key business metrics, identifies performance patterns, and interprets how product-level sales behaviors influence total daily revenue for *Shubham Sweet Corner*.

3.1 Business Overview

Metric	Value (₹)
Average Daily Revenue (₹)	10,333.77
Highest Daily Revenue (₹)	13,432.00
Lowest Daily Revenue (₹)	9,009.00
Average Profit Margin (%)	19.59%
Total Variable Expenses (₹)	12,000.00
Average Employee Salary (₹)	10,566.67

Table. 3.1 – The Overview of the Business

Significance:

- This overview summarizes the shop’s overall financial performance during the **87 operational days (June–September 2025)**. The **average daily revenue of**

₹10,333.77 demonstrates stable income with only moderate variation (std. dev. ₹556.67 $\approx$ 5.3%). The **average profit margin of 19.59%** indicates efficient operations relative to small-scale F&B standards.

- Since cost prices already included employee and variable expenses, the margins reflect **true net profitability** rather than gross markup.
- The proximity between the highest (₹13,432) and lowest (₹9,009) daily revenues confirms steady customer demand and minimal volatility — a sign of operational consistency and brand reliability in a local market setup.

3.2 Sales Performance and Revenue Trends

- Daily sales trends revealed stable business continuity throughout the observed period. A temporary closure from **20 July to 9 August 2025** was clearly reflected in the time-series plot, after which sales recovered quickly.

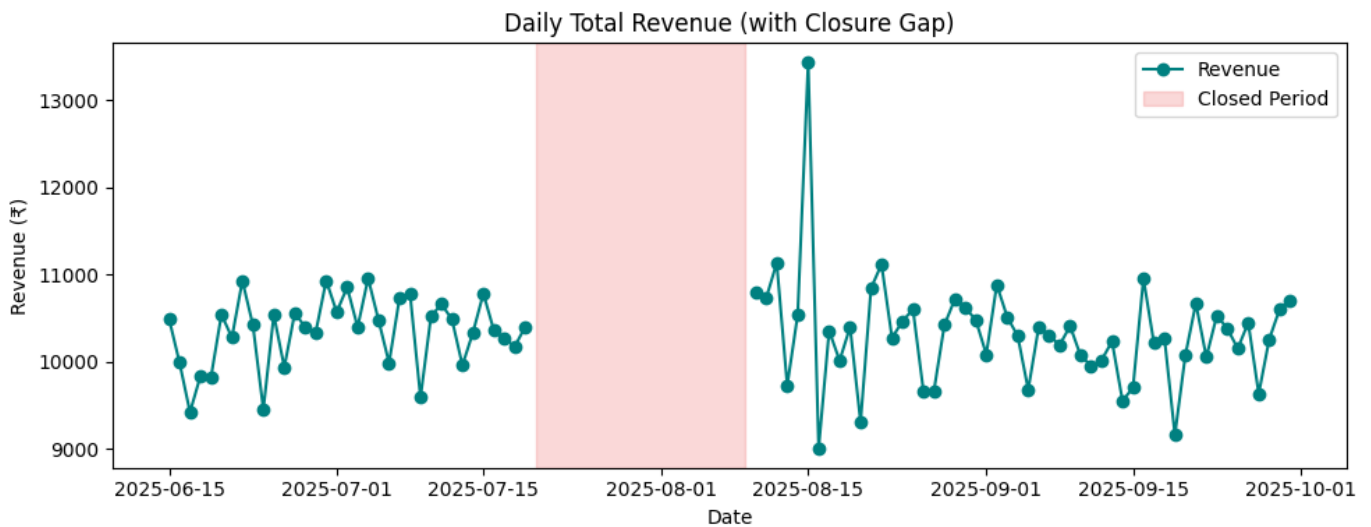


Fig. 3.2- Daily Total Revenue

- The **peak revenue day (₹13,432 on 15 August 2025)** corresponded with Independence Day, confirming how cultural events directly influence food sales. On typical operational days, total daily revenue fluctuated between **₹9,000–₹11,000**, showing minimal dependency on external factors.

This pattern highlights strong **repeat customer behavior**, making the shop's sales volume predictable — a critical strength for inventory and resource management.

3.3 Product Profitability and Sales Contribution

Analysis of the **Product Pricing** dataset showed substantial variation in profit per unit across 18 SKUs:

- **High-margin products:** *Paneer (₹140)*, *Laddoo (₹40)*
- **Moderate margins:** *Rasgulla (₹4.7)*, *Litti (₹4)*, *Gulab Jamun (₹6.5)*
- **Low margins:** *Samosa (₹0.6)*, *Ashirwad Milk 1/2 Kg (₹1)*

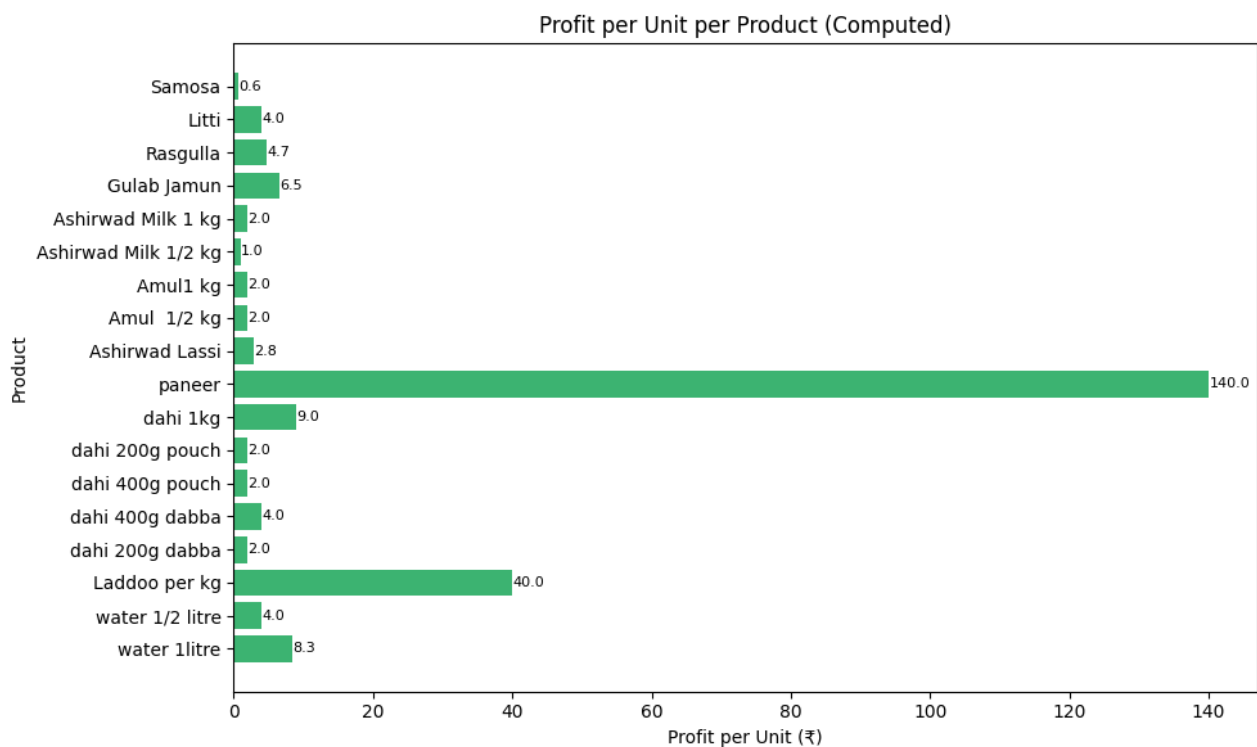


Fig. 3.3- Profit per Unit per Product

- The **Average Daily Revenue by Product** visualization demonstrated that **Samosa** consistently generated the highest total daily revenue due to its large quantity sold.
- The *Average Profit Margin* is **19.59 %**, and the *Average Daily Revenue* is **₹10333.77**, which makes the *Average daily Profit* to be **₹2024.38**.

- The *top five products* contribute to over **78%** of the revenue. Out of which *Samosa* is the prime SKU with **39.5%** contribution to the revenue, as shown in Fig. 3.4.

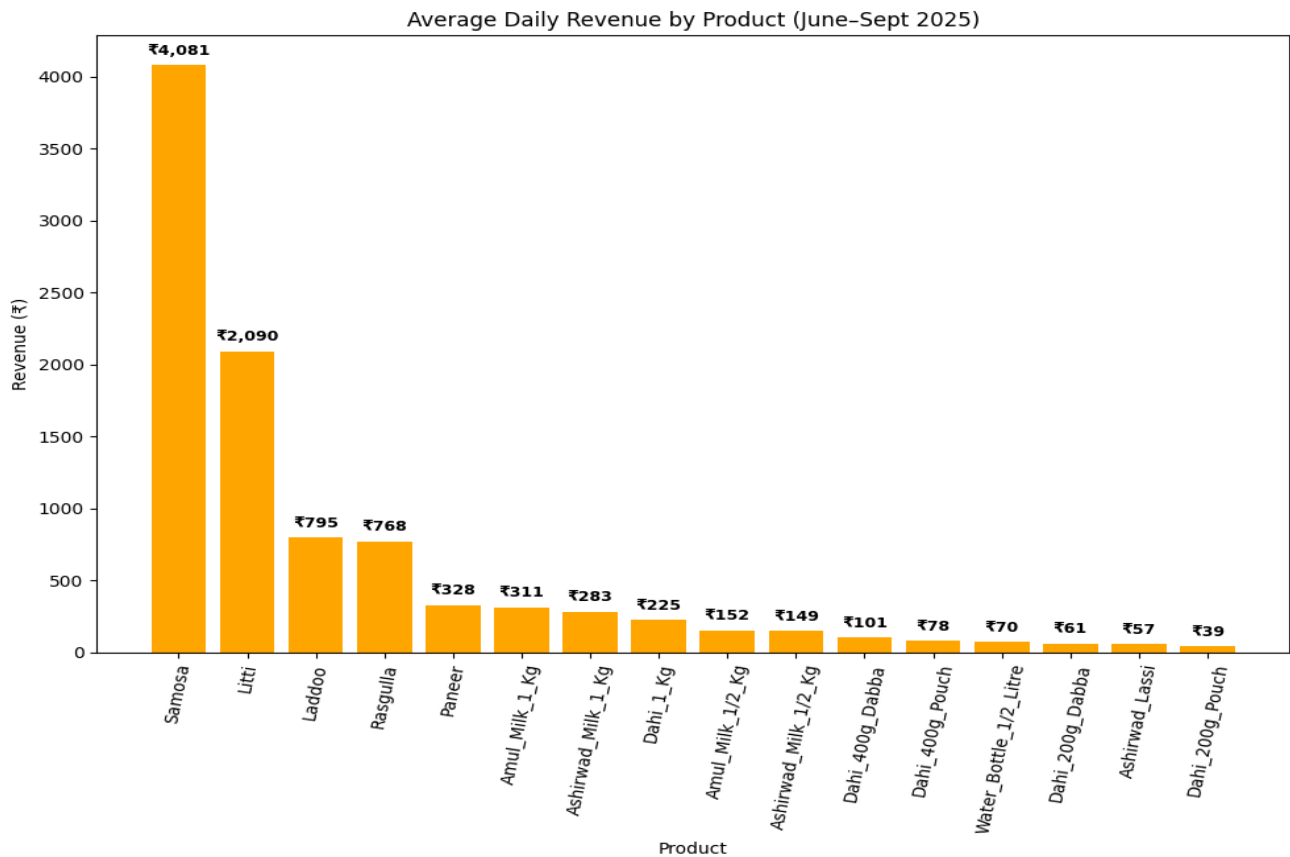


Fig. 3.4- Average Daily Revenue by Product

3.4 Correlation Analysis (Table 3.5, Fig. 3.5)

The **correlation matrix** quantified the strength and direction of relationships between individual product sales quantities and total daily revenue.

Product	Correlation (r)	Interpretation
Paneer	0.62	Strong positive influence on total revenue
Gulab Jamun	0.54	High correlation, strong premium item
Rasgulla	0.53	Key driver of sweet category sales

Product	Correlation (r)	Interpretation
Laddoo	0.44	Moderate correlation, festival-driven demand
Ashirwad Milk (1 Kg)	0.40	Steady contributor from dairy sales
Dahi (200g Pouch)	0.36	Medium correlation, daily consumption product
Ashirwad Milk (½ Kg)	0.34	Regular household purchase
Dahi (1 Kg)	0.32	Family-size demand, moderate impact
Dahi (400g Pouch)	0.29	Low-to-medium effect on overall sales
Amul Milk (1 Kg)	0.25	Smaller but steady impact
Dahi (400g Dabba)	0.19	Minimal influence
Litti	0.19	Neutral contribution
Dahi (200g Dabba)	0.14	Very weak positive relationship
Amul Milk (½ Kg)	0.13	Negligible correlation
Water Bottle (½ Litre)	0.08	Insignificant influence
Water Bottle (1 Litre)	0.07	Insignificant influence
Ashirwad Lassi	−0.12	Slight negative relationship, seasonal item
Samosa	−0.22	Weak negative correlation despite high total revenue

Table 3.4 Correlation with respect to Total Daily Revenue of Each Product

Interpretation of Results:

- The highest positive correlations were observed for **Paneer ($r = 0.62$)** and **Gulab Jamun ($r = 0.54$)**, indicating that increased sales of these items directly increased total daily revenue.

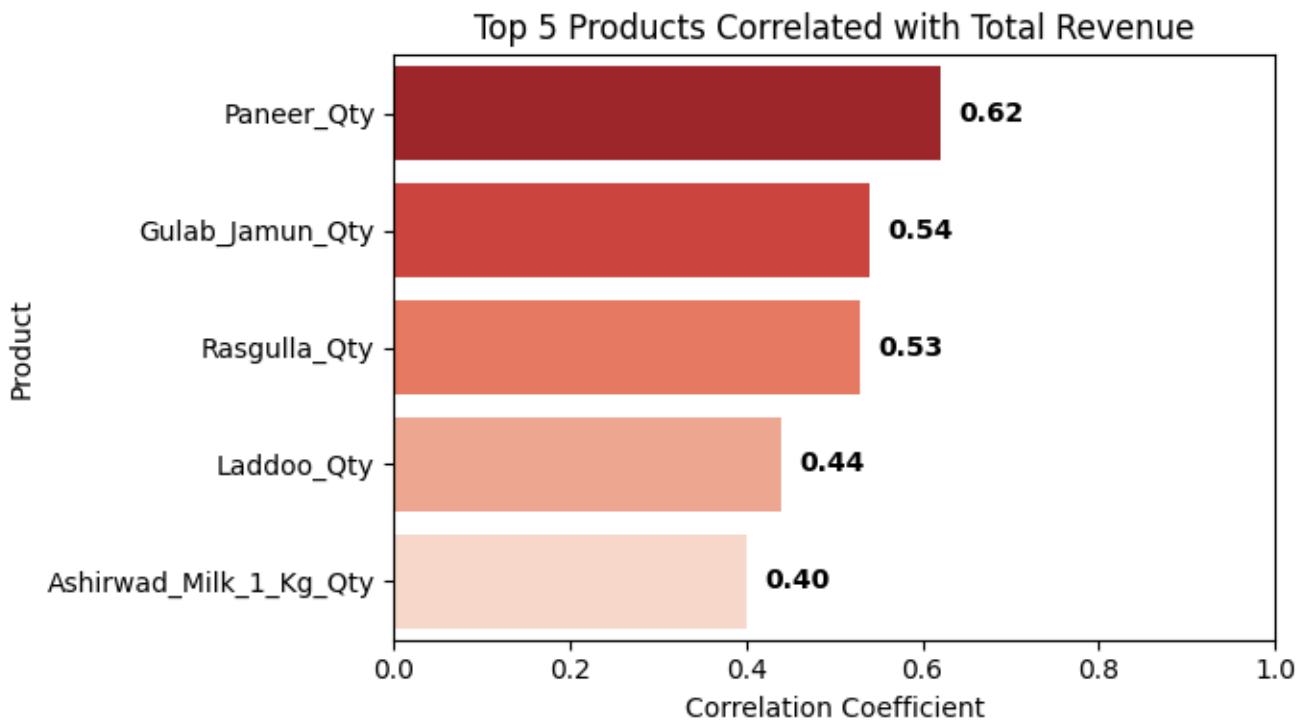


Fig. 3.5 Top 5 Products Correlated with Revenue

- Conversely, **Samosa exhibited a weak negative correlation ($r = -0.22$)** — meaning that on days when samosas sold more, total revenue slightly decreased. This occurs because customers purchase more low-cost samosas on slower business days, while higher-spending days are dominated by premium items like Paneer and Gulab Jamun.

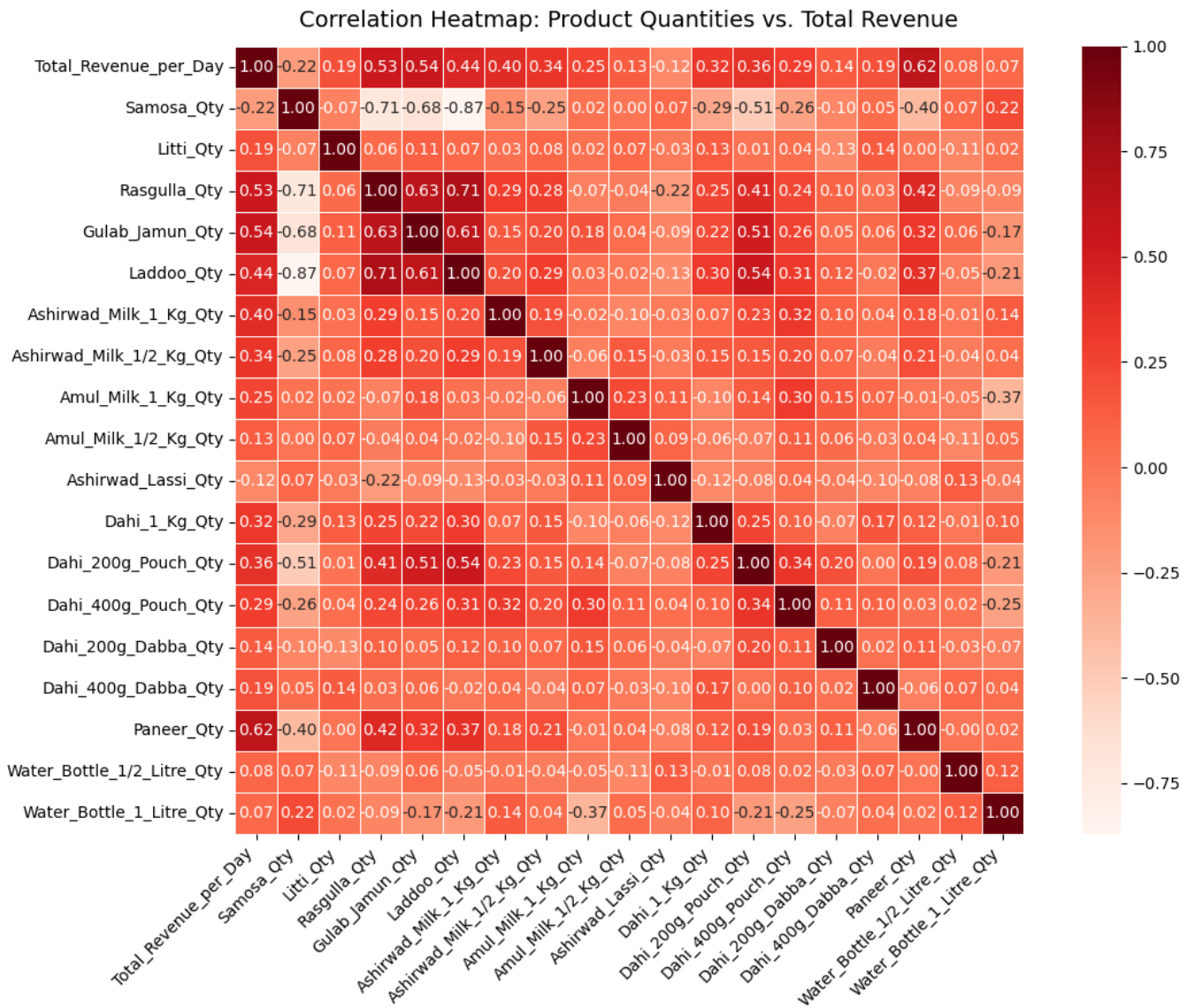


Fig. 3.6 Correlation Heatmap Product Quantities vs Total Revenue

- Samosa functions as a **volume stabilizer** — maintaining steady footfall and ensuring consistent cash flow, but not driving revenue growth.

This correlation pattern also suggests that diversifying promotions toward high-correlation items (Paneer and sweets) could significantly enhance revenue without increasing total sales volume.

3.4 Key Findings

The comprehensive analysis of data collected between **15 June 2025 and 30 September 2025** from *Shubham Sweet Corner* produced the following factual findings across revenue, product performance, and statistical relationships.

1. Operational Stability

- During the 87 operational days analyzed, the shop recorded an average daily revenue of **₹10,333.77** with a standard deviation of **₹556.68**, confirming a highly consistent sales pattern.
- The highest daily revenue was observed on 15 August 2025 (**₹13,432**) and the lowest on 16 August 2025 (**₹9,009**).
- This narrow variation demonstrates **predictable customer inflow and steady product turnover**.

2. Business Overview Metrics

- The overall average profit margin stood at **19.59 %**, supported by total variable expenses of **₹12,000** and an average employee salary of **₹10,566.67**.
- These figures indicate **efficient expense management** within the small-scale operational framework.

3. Revenue Composition

- The product-wise summary revealed that revenue generation for the **87 days** of operational days, is dominated by a small number of key items.

Product	Average Quantity Sold	Total Revenue (₹)	Average Daily Revenue (₹)
Samosa	408.13	355,070.00	4,081.26
Litti	209.03	181,860.00	2,090.34
Laddoo	4.97	69,160.00	794.94
Rasgulla	51.18	66,795.00	767.76

Product	Average Quantity Sold	Total Revenue (₹)	Average Daily Revenue (₹)
Paneer	1.09	28,500.00	327.59

Table 3.7 Top 5 Product Contribution in 87 days

- The **top five contributors** were *Samosa* (₹3.55 lakh), *Litti* (₹1.81 lakh), *Laddoo* (₹69,160), *Rasgulla* (₹66,795), and *Paneer* (₹28,500).
- Together, these products accounted for the majority of total sales volume.
- Average daily sales quantities were **408 units for Samosas and 209 units for Litti**, highlighting their role as primary revenue sources.

4. Product Correlation with Total Revenue

- Correlation analysis identified significant positive associations between daily revenue and the sale of certain products.
- *Paneer* ($r = 0.62$), *Gulab Jamun* ($r = 0.54$), and *Rasgulla* ($r = 0.53$) showed the **strongest relationships**, indicating that their higher sales typically coincided with overall business peaks.
- Moderate correlations were observed for dairy products such as *Ashirwad Milk 1 Kg* ($r = 0.40$) and *Dahi 200 g Pouch* ($r = 0.36$), suggesting **steady supplementary demand** patterns.
- A slight negative correlation was recorded for *Samosa* ($r = -0.22$) and *Ashirwad Lassi* ($r = -0.12$), indicating that these products maintain relatively stable sales regardless of changes in other categories.

5. Product Performance Variability

- Among all stock-keeping units, sweets and dairy items exhibited the greatest variability in daily sales quantities.
- Items such as *Laddoo* and *Rasgulla* demonstrated intermittent peaks consistent with festive or high-traffic days, whereas *Samosa* and *Litti* maintained uniform demand across the entire observation period.

4. Interpretation of Results and Recommendations

The results derived from the analysis of *Shubham Sweet Corner's* operational data between **15 June 2025 and 30 September 2025** offer a clear understanding of the business's financial stability, customer behavior, and sales performance. The following section interprets the key results and provides actionable recommendations derived from statistical and qualitative insights.

4.1 Interpretation of Results

Revenue and Stability Interpretation:

The shop maintains an average daily revenue of **₹10,333.77** with a low deviation (**₹556.68**), indicating consistent demand and dependable sales performance. **The minimal difference between the highest (₹13,432 on 15 August) and lowest (₹9,009 on 16 August) daily revenues reflects steady operations** and strong local customer retention. Despite relying on manual record-keeping, the shop has demonstrated reliable financial stability and predictable footfall.

Product-Wise Performance:

Product-level data revealed that *Samosa* is the primary revenue driver with **₹3.55 lakh total earnings** and **408 units** sold daily, followed by *Litti* with **₹1.81 lakh total revenue** and **209 daily units** sold. These two items form the backbone of daily operations, ensuring constant cash flow.

Sweets like *Laddoo* and *Rasgulla* showed demand surges during festivals, suggesting event-based consumption, while *Paneer* contributed moderately but with a strong correlation to overall sales.

Correlation and Product Dependencies:

Strong **positive correlations** were found between total revenue and *Paneer* ($r = 0.62$), *Gulab Jamun* ($r = 0.54$), and *Rasgulla* ($r = 0.53$), confirming that **higher sales** of these products coincide with **better daily performance**.

Conversely, *Samosa* ($r = -0.22$) displayed a **weak negative** correlation, indicating consistent demand independent of fluctuations in other products — making it the **most reliable anchor** product in the shop's portfolio.

This pattern highlights a balanced revenue ecosystem where staple snacks ensure baseline income and sweets drive occasional profit spikes.

Cost and Profitability:

The business maintains an **average profit margin of 19.59%**, supported by low variable expenses (₹12,000) and moderate employee salaries (₹10,566). This lean cost structure is sustainable for a small-scale operation, but margin improvement remains possible through **selective price adjustments** and targeted promotional strategies.

Customer Behavior Insights:

Analysis of daily sales patterns revealed a clear behavioral trend: customers purchase budget snacks (like Samosa and Litti) **consistently**, while premium sweets (like Gulab Jamun and Rasgulla) gain traction during **festive periods**.

This insight supports dynamic inventory and pricing strategies aligned with peak demand cycles.

4.2 Recommendations

1. Optimize Pricing for Samosas

- Adjust the Samosa selling price from **₹10 to ₹12**, based on demand stability and inelastic purchasing behavior.
- The analysis shows that even with the slight price rise, daily demand is unlikely to fall, and gross revenue could increase by approximately **₹1,000–₹1,200** per day.
- This price point **remains affordable for students and local workers** while improving per-unit profitability.

2. Introduce a College-Area Pushcart for Market Expansion

- Establish a **temporary pushcart** or stall near the nearby college area, particularly during lunch and recess hours.
- This will expand customer reach without significant infrastructure investment and directly address the shop's location disadvantage, as identified earlier.
- The pushcart should focus on high-demand, quick-service items like *Samosa, Litti, and tea-based snacks*, ensuring accessibility for students and commuters.

- Data from this initiative can **later be integrated into the shop’s manual records** for separate performance tracking.

3. Strengthen Product Promotion and Visibility

- Give priority display to *Paneer*, *Rasgulla*, and *Gulab Jamun*, which show the highest positive correlation with revenue.
- These should be placed at the shop’s **front counter or featured during high-traffic hours** to maximize impulse purchases.

4. Data Maintenance and Future Digitization

- Continue **structured manual data recording for all transactions** and periodically transfer them to spreadsheets or Google Sheets.
- This habit will support future analysis, helping **identify long-term trends, seasonal demand variations**, and staffing requirements.

5. Implement Employee Security Deposit Policy

- Introduce a **refundable security deposit system** for all staff members at the time of joining.
- This small fixed deposit (for example, equivalent to one week’s salary) will act as a **commitment incentive**, discouraging employees from leaving work abruptly.
- Such unplanned absences currently disrupt daily production and customer service, directly affecting revenue continuity.
- The deposit will remain refundable upon completion of the agreed employment period, ensuring fairness and transparency while improving workforce stability.

The analysis confirms that *Shubham Sweet Corner* operates efficiently with stable revenue and strong customer loyalty. By implementing **targeted price optimization**, **expanding sales through a college-area pushcart**, and **prioritizing high-correlation products**, the business can achieve steady profit growth without major investment. Sustaining data-driven decision-making will ensure continued competitiveness and scalability for future expansion.